



## Modelling Financial Markets

### Seminar 20: Path Signatures and Realised Volatility — Hands-on



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## Acronyms used in this material (1/2)

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ AUDUSD: Australian dollar / US dollar exchange rate (FX pair code)
- ▣ BH: Benjamini–Hochberg (procedure)
- ▣ BIC: Bayesian Information Criterion
- ▣ CI: Confidence Interval
- ▣ CL: Crude oil (WTI light sweet) futures, NYMEX ticker
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ CVX: Chevron Corporation (NYSE ticker)
- ▣ DM: Diebold–Mariano (test)
- ▣ ES: E-mini S&P 500 futures (CME ticker)
- ▣ EURUSD: Euro / US dollar exchange rate (FX pair code)
- ▣ FDR: False Discovery Rate
- ▣ FWER: Family-Wise Error Rate
- ▣ FX: Foreign eXchange
- ▣ GBPUSD: British pound / US dollar exchange rate (FX pair code)
- ▣ GW: Giacomini–White (test)
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ HAR: Heterogeneous AutoRegressive (model)
- ▣ HARQ: HAR model with Realised Quarticity
- ▣ JPM: JPMorgan Chase & Co. (NYSE ticker)
- ▣ LASSO: Least Absolute Shrinkage and Selection Operator
- ▣ OLS: Ordinary Least Squares
- ▣ OPEC: Organization of the Petroleum Exporting Countries
- ▣ PRDS: Positive Regression Dependence on a Subset (of p-values)
- ▣ QLIKE: Quasi-LIKElihood (loss function)

## Acronyms used in this material (2/2)

- ▣ RV: Realised Variance
- ▣ SHAR: Semivariance HAR model
- ▣ SHW: Sherwin-Williams Company (NYSE ticker)
- ▣ USDCAD: US dollar / Canadian dollar exchange rate (FX pair code)
- ▣ USDJPY: US dollar / Japanese yen exchange rate (FX pair code)
- ▣ VOLARE: VOLatility Archive for Realized Estimates (database)

## Today's Question and Required Outputs

- Question of the seminar: **does the shape of the recent volatility path help to forecast realised variance, and how do we test that honestly?**
- Route in class (about  $2 \times 50$  minutes of work, the rest discussion)
  - ▶ A1, A7, A9 on paper: a signature step by step, the QLIKE loss, kernel weights
  - ▶ B1, B3 on JPM: the Lévy area as a predictor and the HAR family, with HAC errors
  - ▶ B5: Sig-LK against log-HAR with DM, GW and a bootstrap interval; B7: Holm and BH over 50 assets
  - ▶ C2: find the errors in an AI answer
- Required outputs (checklist on the last core slide): the A answers, one figure and one table for each of B1, B3, B5, B7, the corrected C2 code, one interpretation sentence per exercise
- Notebook for today: [open the seminar notebook in Colab](#); background: Lecture 20 and Chapter 9
- Seminar: Antoaneta Amza, PhD candidate, [antoaneta.amza@csie.ase.ro](mailto:antoaneta.amza@csie.ase.ro)

## Exercise Map

| Exercise        | Question                                                         | Type     | Model              |
|-----------------|------------------------------------------------------------------|----------|--------------------|
| A1              | how does a signature record the order of moves?                  | Solved   | –                  |
| A7              | which forecast minimises QLIKE, which errors cost more?          | Solved   | –                  |
| A9              | how does the temperature trade similarity for sample size?       | Solved   | –                  |
| B1              | does the Lévy area predict next month's RV of JPM?               | Solved   | –                  |
| B3              | do the SHAR and HARQ effects hold for JPM?                       | Solved   | –                  |
| B5              | does Sig-LK beat log-HAR for JPM, one day ahead?                 | Solved   | –                  |
| B7              | how many wins survive Holm and BH over 50 assets?                | Solved   | –                  |
| C2              | what is wrong in an AI answer?                                   | Proposed | A1, A7, B5         |
| Extension packs | A2–A6, A8, A10, B2, B4, B6, B8, C1 (optional, after the seminar) | both     | named on each task |

- **[Solved]**: full solution in the slides and in the notebook, a model to copy; **[Proposed]**: you solve it, the solution is discussed in class
- Part A: derivations on paper; Part B: verification and inference on VOLARE data, each ending with an interpretation question; Part C: open questions and AI critique

## Data Manifest

| File                          | Symbols                                | Sample                    | Used in            |
|-------------------------------|----------------------------------------|---------------------------|--------------------|
| realized_variance_stocks.csv  | JPM (all 40 stocks in C1)              | 2 Jan 2015 – 30 Jan 2026  | A5, B1, B3, B5, C1 |
| realized_variance_futures.csv | ES (E-mini S&P 500),<br>CL (crude oil) | 28 Sep 2009 – 30 Jan 2026 | B2, B4, B6         |
| ch20_results.json             | 50 assets, results of<br>Lecture 20    | one-sided DM $p$ -values  | B7, B8             |

- Columns of the VOLARE files [Cipollini et al. \(2026\)](#), one row per asset and trading day
  - ▶ rv5: realised variance from five-minute returns,  $RV_t = \sum_i r_{t,i}^2$ ; rsp5, rsn5: its positive and negative parts (semivariances)
  - ▶ rq5: realised quarticity, the scale of the measurement error of RV; open\_price, close\_price: for the open-to-close log return  $r_t$
- VOLARE is free for research at [volare.unime.it](http://volare.unime.it): download the three realised-variance files and set their folder in the first notebook cell

## Setup: Run and Verify

```
# notebook section "Signatures, models and tests"
# defines load_asset(kind, symbol): columns
# rv, rsp, rsn, rq and r = ln(close/open)
a = load_asset('stocks', 'JPM')
print(len(a), a.index[0].date(), a.index[-1].date())
print(pd.DataFrame({'RV_x_1e4': 1e4 * a.rv,
                    'r(%)': 100 * a.r}).head(3))
```

- Expected output: 2786 trading days, 2015-01-02 to 2026-01-30

| date       | RV $\times 10^4$ | r (%) |
|------------|------------------|-------|
| 2015-01-02 | 1.463            | -0.22 |
| 2015-01-05 | 1.855            | -2.56 |
| 2015-01-06 | 2.974            | -2.77 |

- RV of  $1.46 \times 10^{-4}$  is a daily volatility of  $\sqrt{1.46 \times 10^{-4}} = 1.2\%$
- If your count or first value differs, check the folder, the file and the symbol

## Notation Used in Every Exercise

- $RV_t$ : five-minute realised variance of day  $t$ ;  $r_t = \ln(\text{close}_t / \text{open}_t)$ : open-to-close log return
- HAR averages:  $RV_t^{(w)} = \frac{1}{5} \sum_{j=0}^4 RV_{t-j}$ ,  $RV_t^{(m)} = \frac{1}{22} \sum_{j=0}^{21} RV_{t-j}$ ; log-HAR regresses  $\ln RV_{t+1:t+h}$  on  $\ln RV_t$ ,  $\ln RV_t^{(w)}$ ,  $\ln RV_t^{(m)}$ , with  $RV_{t+1:t+h}$  the mean RV over the next  $h$  days
- Path  $X$  with increments  $\Delta_k$ ; signature levels  $S^{(1)} = \sum_k \Delta_k$  and  $S^{(2)}$  (a  $d \times d$  matrix); Lévy area  $A^{12} = (S^{12} - S^{21})/2$
- Loss  $\text{QLIKE}(RV, F) = RV/F - \ln(RV/F) - 1$  for a forecast  $F$ ; loss difference  $d_t = \text{QLIKE}_t^{(1)} - \text{QLIKE}_t^{(2)}$ ; DM statistic  $t = \bar{d} / \sqrt{\hat{\Omega}_{\text{NW}}/T}$  with a Newey–West (HAC) long-run variance
- Kernel weights  $w_\tau = e^{-\gamma \delta_\tau} / \sum_s e^{-\gamma \delta_s}$  for signature distances  $\delta_\tau$ ; Kish effective sample size  $n_{\text{eff}} = 1 / \sum_\tau w_\tau^2$
- One worked line:  $RV = 1$ ,  $F = 2$ :  $\text{QLIKE} = 0.5 - \ln 0.5 - 1 = 0.193$

## A1 [Solved]: A Depth-2 Signature, Step by Step — Task

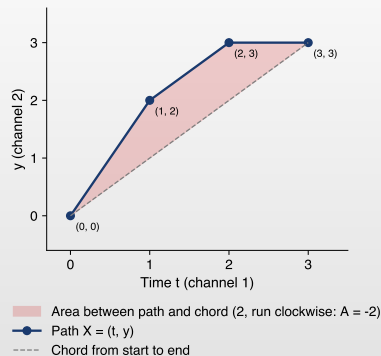
- **Question:** how does a depth-2 signature record *when* a series moved, not only how much?
- **Inputs**
  - ▶ A series takes the values  $y = (0, 2, 3, 3)$  on days  $t = 0, 1, 2, 3$ ; no data file
  - ▶ The time-augmented path  $X_t = (t, y_t)$  is linear between consecutive days (channel 1 = time, channel 2 =  $y$ )
- **Tasks**
  - (a) Write the three increments  $\Delta_1, \Delta_2, \Delta_3 \in \mathbb{R}^2$  and compute level 1,  $S^{(1)} = \sum_k \Delta_k$ .
  - (b) Use Chen's identity to show that  $S^{(2)} = \sum_k \frac{1}{2} \Delta_k \Delta_k^\top + \sum_{j < k} \Delta_j \Delta_k^\top$ , then compute this  $2 \times 2$  matrix.
  - (c) Check the result with the shuffle identity  $S^i S^j = S^{ij} + S^{ji}$  for  $i = j = 1$  and for  $i = 1, j = 2$ .
  - (d) Compute the Lévy area  $A^{12} = (S^{12} - S^{21})/2$  and draw the path with its chord.
- **Report:**  $S^{(1)}$ ,  $S^{(2)}$  and  $A^{12}$
- **Interpretation:** what does the sign of  $A^{12}$  say about when the series rose?
- Notebook: section A1

## A1 [Solved]: Solution (1/2) — Levels 1 and 2

- (a) Increments:  $\Delta_1 = (1, 2)$ ,  $\Delta_2 = (1, 1)$ ,  $\Delta_3 = (1, 0)$ ; level 1:  $S^{(1)} = (3, 3)$ , the total change in time and in  $y$
- (b) One segment has  $\text{Sig} = (1, \Delta, \frac{1}{2}\Delta\Delta^\top, \dots)$ ; Chen multiplies the segments, and level 2 of the product collects every square term once and every ordered pair  $j < k$  once
- Squares:  $\frac{1}{2} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1.5 & 1.5 \\ 1.5 & 2.5 \end{bmatrix}$
- Ordered pairs:  $\Delta_1\Delta_2^\top + \Delta_1\Delta_3^\top + \Delta_2\Delta_3^\top = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 2 & 0 \end{bmatrix} + \begin{bmatrix} 1 & 0 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$
- Sum:  $S^{(2)} = \begin{bmatrix} 4.5 & 2.5 \\ 6.5 & 4.5 \end{bmatrix}$

## A1 [Solved]: Solution (2/2) — Shuffle Check and Lévy Area

- (c) Shuffle,  $i = j = 1$ :  $S^1 S^1 = 9 = 2S^{11} = 2 \cdot 4.5$ ;  $i = 1, j = 2$ :  $S^1 S^2 = 9 = S^{12} + S^{21} = 2.5 + 6.5$ : both hold
- (d) Lévy area:  $A^{12} = (2.5 - 6.5)/2 = -2$
- Geometric check (chart): the path lies above its chord  $y = t$ ; the shaded polygon has area 2 and is run clockwise, hence the minus sign
- Interpretation: with channels (time,  $y$ ), a negative area means that  $y$  rose *early* in the window; the same net change spread evenly would give  $A^{12} = 0$

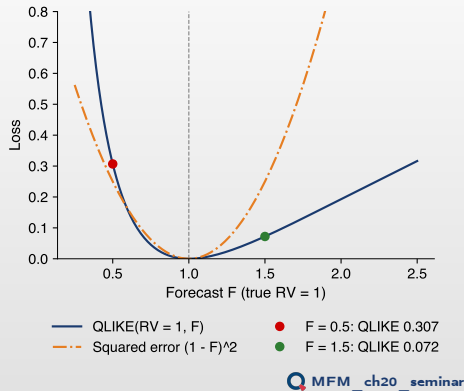


## A7 [Solved]: QLIKE: Minimiser and Asymmetry — Task

- **Question:** which forecast minimises expected QLIKE, and which errors does QLIKE punish more?
- **Inputs:**  $\text{QLIKE}(RV, F) = RV/F - \ln(RV/F) - 1$  with realised proxy  $RV > 0$  and forecast  $F > 0$ ;  $\mathbb{E}_t =$  expectation given the information at the forecast origin
- **Tasks**
  - (a) Differentiate  $\mathbb{E}_t[\text{QLIKE}(RV, F)]$  with respect to  $F$  and solve the first-order condition.
  - (b) Check the second-order condition.
  - (c) For  $RV = 1$ , compute QLIKE and the squared error for  $F = 0.5$  and for  $F = 1.5$ , and plot both losses against  $F$ .
- **Report:** the minimiser, the four losses and the chart
- **Interpretation:** which error does QLIKE punish more, and why does this suit risk management?
- **Notebook:** section A7

## A7 [Solved]: Solution

- (a)  $\mathbb{E}_t[\text{QLIKE}] = \mathbb{E}_t[\text{RV}]/F + \ln F - \mathbb{E}_t[\ln \text{RV}] - 1$ ;  
 $\partial/\partial F = -\mathbb{E}_t[\text{RV}]/F^2 + 1/F = 0 \Rightarrow F^* = \mathbb{E}_t[\text{RV}]$
- (b)  $\partial^2/\partial F^2 = 2\mathbb{E}_t[\text{RV}]/F^3 - 1/F^2 = 1/F^{*2} > 0$  at  $F^*$ : a minimum;  $F^*$  depends only on  $\mathbb{E}_t[\text{RV}]$ , so with a conditionally unbiased proxy the *expected-loss* ranking is robust [Patton \(2011\)](#)
- (c)  $\text{QLIKE}(1, 0.5) = 2 - \ln 2 - 1 = 0.307$ ;  
 $\text{QLIKE}(1, 1.5) = 0.667 + \ln 1.5 - 1 = 0.072$ ; squared errors 0.25 and 0.25
- Interpretation: QLIKE punishes under-prediction of variance about four times more than the same over-prediction; under-predicted risk is the costly error (too little capital, breached VaR)

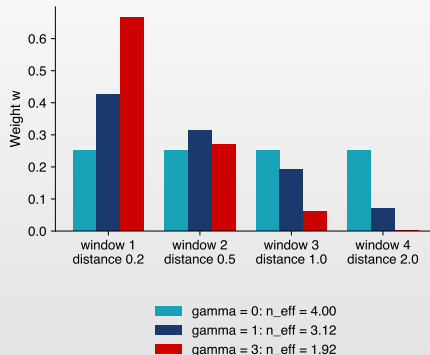


## A9 [Solved]: Kernel Weights and Effective Sample Size — Task

- **Question:** how does the kernel temperature  $\gamma$  trade similarity against the number of observations?
- **Inputs**
  - ▶ Four past windows with signature distances  $\delta = (0.2, 0.5, 1.0, 2.0)$  to the current window
  - ▶ Weights of Gu et al. (Eq. 12):  $w_\tau = e^{-\gamma\delta_\tau} / \sum_s e^{-\gamma\delta_s}$ ; Kish effective sample size  $n_{\text{eff}} = 1 / \sum_\tau w_\tau^2$
- **Tasks**
  - (a) Compute the four weights for  $\gamma = 1$ .
  - (b) Compute the four weights for  $\gamma = 3$ .
  - (c) Compute  $n_{\text{eff}}$  in both cases and compare with equal weights ( $\gamma = 0$ ).
- **Report:** two weight vectors, two  $n_{\text{eff}}$ , a bar chart
- **Interpretation:** what is gained and what is lost when  $\gamma$  increases?
- **Notebook:** section A9

## A9 [Solved]: Solution

- (a)  $e^{-\delta} = (0.819, 0.607, 0.368, 0.135)$ , sum 1.929:  
 $w = (0.425, 0.315, 0.191, 0.070)$
- (b)  $e^{-3\delta} = (0.549, 0.223, 0.050, 0.002)$ , sum 0.824:  
 $w = (0.666, 0.271, 0.060, 0.003)$
- (c)  $n_{\text{eff}} = 4$  for  $\gamma = 0$ , 3.12 for  $\gamma = 1$  and 1.92 for  $\gamma = 3$
- Interpretation: a higher temperature focuses on the most similar window (less bias from dissimilar periods) but halves the information (more variance); in the lecture  $n_{\text{eff}} \approx 816$  of 1000: little is lost, and little changes



## B1 [Solved]: Does the Lévy Area Predict Volatility? (JPM) — Task

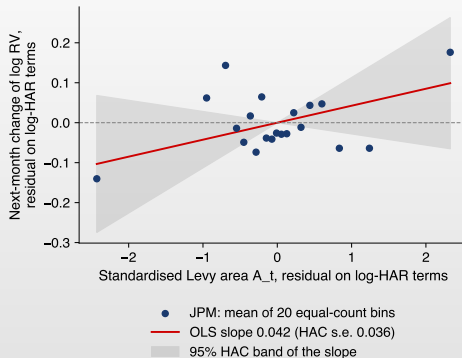
- **Question:** does the order of price and volatility moves over the last month predict JPM's volatility next month, beyond log-HAR?
- **Inputs**
  - ▶ VOLARE [Cipollini et al. \(2026\)](#), JPM (JPMorgan Chase), 2 Jan 2015 – 30 Jan 2026, trading days only; columns `rv5`, `open_price`, `close_price`
  - ▶ Regressor:  $A_t$  = Lévy area of the 22-day path (cumulative  $r$ ,  $\ln RV$ ) ending at day  $t$ , divided by its standard deviation
  - ▶ Target:  $y_t = \ln RV_{t+1:t+22} - \ln RV_t^{(m)}$ , next month's log RV minus the past month's (consecutive targets overlap in 21 of 22 days)
- **Tasks**
  - (a) For every day, build the path of the last 22 days, compute its depth-2 signature and  $A_t = (S^{12} - S^{21})/2$ , then standardise  $A_t$ .
  - (b) Regress  $y_t$  by OLS on a constant, the three log-HAR terms and  $A_t$ , with Newey–West standard errors and 22 lags.
  - (c) Test  $H_0$ : coefficient of  $A_t = 0$  at 5%, two-sided, and compare the  $R^2$  with and without  $A_t$ .
  - (d) Draw a binned scatter of  $y_t$  against  $A_t$ , both residualised on the log-HAR terms.
- **Report:** coefficient, HAC s.e.,  $t$ , decision, both  $R^2$ , the chart
- **Interpretation:** why 22 lags, and is the gain economically relevant?
- Notebook: section B1

## B1 [Solved]: Solution (1/2) — Steps

- (a) Signature of each 22-point window with channels (return,  $\ln RV$ );  $A_t$  is negative in 46% of the windows (compatible with, but not identifying, leverage-type sequences)
- (b) 2743 usable days; coefficient of  $A_t = +0.042$ , HAC s.e. 0.036
- (c)  $t = +0.042/0.036 = +1.18$ ;  $|t| < 1.96$ : do not reject  $H_0$  at 5%;  $R^2$ : 28.0% without  $A_t$ , 28.6% with  $A_t$
- Why 22 lags: consecutive targets share 21 of their 22 days, so the errors are serially dependent even under  $H_0$ ; persistent RV can extend the dependence, so a check with 44 lags is advisable

## B1 [Solved]: Solution (2/2) — Chart and Interpretation

- (d) Chart: bins scatter around zero; the HAC band of the slope contains the horizontal line
- Relevance: half a point of  $R^2$  next to the 28% explained by the log-HAR terms is negligible
- Interpretation: for JPM the order of moves between price and volatility adds no reliable information about next month's volatility



## B3 [Solved]: HAR, SHAR and HARQ for JPM — Task

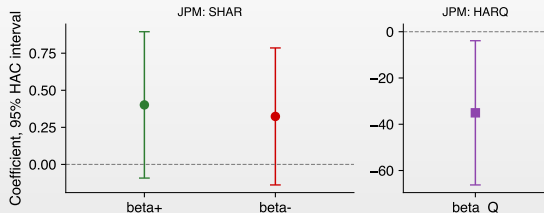
- **Question:** do bad volatility (SHAR) and the measurement error of RV (HARQ) change next-day forecasts for JPM?
- **Inputs**
  - ▶ VOLARE [Cipollini et al. \(2026\)](#), JPM (JPMorgan Chase), 2 Jan 2015 – 30 Jan 2026, trading days only; columns rv5, semivariances rsp5, rsn5, quarticity rq5; target: next-day RV in levels, full sample
  - ▶ SHAR [Patton & Sheppard \(2015\)](#):  $RV_{t+1} = \beta_0 + \beta^+ RV_t^+ + \beta^- RV_t^- + \beta_w RV_t^{(w)} + \beta_m RV_t^{(m)} + \varepsilon_{t+1}$
  - ▶ HARQ [Bollerslev, Patton & Quaedvlieg \(2016\)](#): HAR plus the term  $\beta_Q RV_t (RQ_t^{1/2} - \overline{RQ}^{1/2})$ ;  $V =$  Newey–West (5 lags) covariance matrix of the estimates
- **Tasks**
  - (a) Estimate HAR [Corsi \(2009\)](#) by OLS and report the three slopes with HAC standard errors.
  - (b) Estimate SHAR and test  $H_0 : \beta^- = \beta^+$  against  $\beta^- > \beta^+$  with  $t = (\hat{\beta}^- - \hat{\beta}^+) / \sqrt{V_{--} + V_{++} - 2V_{+-}}$ .
  - (c) Estimate HARQ and test  $H_0 : \beta_Q = 0$  against  $\beta_Q < 0$ ; plot  $\beta^+$ ,  $\beta^-$  and  $\beta_Q$  with 95% HAC intervals.
- **Report:** estimates,  $t$ ,  $p$ , decisions, the coefficient chart
- **Interpretation:** do the two effects of the literature hold for this stock?
- Notebook: section B3

### B3 [Solved]: Solution (1/2) — Estimates and Tests

- (a) HAR (2 764 days):  $\beta_d = 0.367$  (0.080),  $\beta_w = 0.494$  (0.149),  $\beta_m = -0.026$  (0.095)
- (b) SHAR:  $\beta^+ = 0.401$  (0.252),  $\beta^- = 0.324$  (0.236); difference  $-0.078$ ,  $t = -0.17$ , one-sided  $p = 0.566$ : do not reject  $H_0$
- (c) HARQ:  $\hat{\beta}_Q = -35.062$  (15.904),  $t = -2.20 < -1.645$ : reject  $H_0$  at 5% one-sided

## B3 [Solved]: Solution (2/2) — Chart and Interpretation

- Chart: the two SHAR intervals are wide and overlap (the semivariances are highly correlated for one stock); the HARQ interval lies below zero
- Interpretation: on noisy days HARQ trusts the day's RV less, and this holds for JPM; the good/bad asymmetry is a panel result [Patton & Sheppard \(2015\)](#), not detectable for JPM alone



- SHAR beta+ (positive semivariance)
- SHAR beta- (negative semivariance)
- HARQ beta\_Q

## B5 [Solved]: Sig-LK vs log-HAR for JPM, One Day Ahead — Task

- **Question:** does the signature method of the lecture (Sig-LK) forecast JPM's next-day RV better than log-HAR?
- **Inputs**
  - ▶ VOLARE [Cipollini et al. \(2026\)](#), JPM (JPMorgan Chase), 2 Jan 2015 – 30 Jan 2026, trading days only; columns `rv5`, `open_price`, `close_price`
  - ▶ Design of the lecture: rolling window of 1000 days, re-estimation every 5 days, path (time,  $\ln RV$ , cumulative  $r$ ) over 22 days, depth 3, two-step LASSO with BIC, kernel temperature  $c = 0.5$
- **Tasks**
  - (a) Produce one-day forecasts of Sig-LK and log-HAR from 27 Jan 2020 to 29 Jan 2026 (1511 days) and compute the mean QLIKE of each.
  - (b) Compute the ratio of the two mean losses and its 95% CI with the stationary bootstrap (mean block length 10, 1000 replications).
  - (c) Run the DM test (HAC variance, bandwidth  $\max(h, \lceil T^{1/3} \rceil)$ ) and the conditional GW test with instruments  $(1, d_{t-1})$  [Giacomini & White \(2006\)](#).
  - (d) Plot the cumulative loss difference  $\sum_{s \leq t} d_s$ .
- **Report:** losses, ratio and CI,  $t$ ,  $p$ , decisions, the chart
- **Interpretation:** is there any evidence that the signature adds value for JPM?
- **Notebook:** section B5

## B5 [Solved]: Solution (1/2) — Losses and Tests

- (a) Mean QLIKE: Sig-LK 0.1726, log-HAR 0.1727; on average 4.2 regressors selected, effective sample size 851 of 1000
- (b) Ratio 1.000; bootstrap 95% CI [0.956, 1.047] contains 1
- (c) DM:  $d_t = \text{QLIKE}_t^{\text{Sig-LK}} - \text{QLIKE}_t^{\text{log-HAR}}$ ,  $t = \bar{d} / \sqrt{\hat{\Omega}_{\text{NW}} / T} = -0.02$ , one-sided  $p = 0.493$ : do not reject equal accuracy
- GW: Wald statistic  $2.85 \sim \chi^2(2)$ ,  $p = 0.241$ : the instruments  $(1, d_{t-1})$  give no significant evidence that the loss difference is predictable

## B5 [Solved]: Solution (2/2) — Cumulative Loss Difference

- (d) Upward segments: log-HAR better; downward segments: Sig-LK better; the largest moves sit in March 2020 and April 2025
- A handful of stress days decides the sign of the total: report the cumulative path, not only the mean
- Interpretation: no evidence; the two forecasts are practically identical, and the CI rules out gains or losses larger than about 5%



— Cumulative QLIKE(Sig-LK) - QLIKE(log-HAR), JPM,  $h = 1$

## B7 [Solved]: Holm and BH, Step by Step — Task

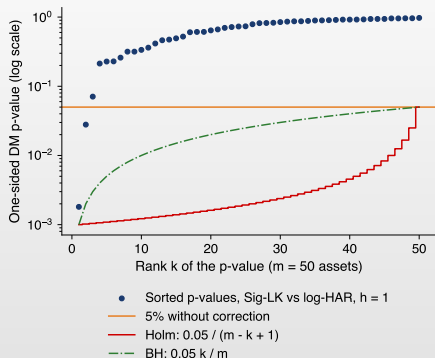
- **Question:** after correcting for 50 tests, on how many assets does Sig-LK beat log-HAR one day ahead?
- **Inputs**
  - ▶ Lecture 20 results (ch20\_results.json): one-sided DM  $p$ -values for “Sig-LK better than log-HAR” at  $h = 1$  (QLIKE), one per asset; family = the 50 assets
  - ▶ The five FX pairs, sorted: EURUSD 0.071, USDJPY 0.213, USDCAD 0.336, AUDUSD 0.718, GBPUSD 0.817
  - ▶ Holm (FWER 5%): compare the  $k$ -th smallest  $p_{(k)}$  with  $0.05/(m - k + 1)$  and stop at the first non-rejection; BH (FDR 5%) [Benjamini & Hochberg \(1995\)](#): reject the  $k$  smallest for the largest  $k$  with  $p_{(k)} \leq 0.05 k/m$
- **Tasks**
  - (a) Apply Holm to the five FX  $p$ -values ( $m = 5$ ).
  - (b) Apply BH to the same five.
  - (c) Repeat both procedures for all 50 assets, count the rejections and plot the sorted  $p$ -values against the three thresholds.
- **Report:** rejections raw, Holm, BH; the chart
- **Interpretation:** what may we claim, and about how many assets?
- Notebook: section B7

## B7 [Solved]: Solution (1/2) — Steps

- (a) Holm,  $m = 5$ :  $p_{(1)} = 0.071$  vs  $0.05/5 = 0.01$ : no rejection, stop; Holm-adjusted  $p$ : 0.354, 0.852, then 1
- (b) BH: thresholds 0.01, 0.02, 0.03, 0.04, 0.05; every  $p_{(k)}$  is above its threshold: no rejection; the smallest BH-adjusted  $p$  is 0.354
- (c) 50 assets: 2 raw rejections; Holm: the smallest, ES with  $p = 0.002$ , is above  $0.05/50 = 0.001$ : stop, 0 rejections; BH:  $p_{(1)} > 0.001$  and  $p_{(2)} = 0.028 > 0.002$ , no  $k$  qualifies: 0

## B7 [Solved]: Solution (2/2) — Chart and Interpretation

- Chart: only the lowest points cross the 5% line; none reaches the Holm staircase or the BH line
- Holm is valid under any dependence; BH needs independent or positively dependent (PRDS)  $p$ -values [Benjamini & Yekutieli \(2001\)](#)
- Interpretation: no asset-level improvement survives either correction; ES is at most a lead for a new, pre-registered study



## C2 [Proposed]: Critique an AI Answer — the Answer

□ **Prompt** sent to an AI assistant: “rv holds the daily 5-minute RV of one stock. Build the signature LASSO of Gu et al. for next-day log RV and evaluate it with QLIKE on the last 30% of the sample.”

□ **Answer:** the code below and the conclusion

```

lrsv = np.log(rv.values)
S = np.array([sig_path(lrv[t-21:t+1, None], depth=3)      # 22-day window
              for t in range(21, len(lrv) - 1)])
y = lrv[22:]                                              # next-day log RV
lam = LassoCV(cv=5).fit(S, y).alpha_                     # choose lambda
cut = int(0.7 * len(y))
m = Lasso(alpha=lam).fit(S[:cut], y[:cut])
f = np.exp(m.predict(S[cut:]))
qlike = lambda a, b: a / b - np.log(a / b) - 1
print(qlike(f, np.exp(y[cut:])).mean())

```

□ **Conclusion of the AI:** “The signature LASSO has a lower QLIKE than HAR on the test set, so path signatures capture the shape of volatility and improve the forecast.”

## C2 [Proposed]: Critique an AI Answer — Task

Solution discussed in class

- **Question:** which errors in the answer would change the conclusion?
- **Model:** A1 (time augmentation), A7 (QLIKE), B5 (evaluation without look-ahead); the code contains **at least three** errors
- **Tasks**
  - (a) Find at least three errors; for each, name the line and explain in one sentence why it is wrong.
  - (b) For each error, describe a check that reveals it without knowing the answer (a unit test, a timing test, a known case).
  - (c) Assess the exponential back-transform and whether the code is the two-step weighted method of Gu et al.
  - (d) Write the corrected code and compare it with log-HAR on the same split, with a DM test.
  - (e) Log each error as in [AI\\_ERRORS.md](#): request, answer, error, how found, correction.
- **Interpretation:** does the conclusion follow even if the code were correct?
- AI code usually runs without an error message: the mistakes are in the method, not in the syntax

## Submission Checklist and Team Repository

### Hand in before the next seminar

- A1, A7, A9: the matrices, losses and weights, with the three charts
- B1, B3, B5, B7: one figure and one table each, as in the solutions
- C2: the list of errors, the checks, the corrected code and its DM test against log-HAR
- One interpretation sentence per exercise

### Team repository

- One Quantlet folder: B5 for a second stock of your choice, with the chart
- Log AI use in `AI_USE.md` and caught errors in `AI_ERRORS.md`
- Extension packs are optional; C1 is a possible project start

## Extension Packs: What Each One Needs

| Pack                             | Exercises                           | Prerequisites                   |
|----------------------------------|-------------------------------------|---------------------------------|
| 1. Signature algebra             | A2, A3 [Solved], A4                 | A1                              |
| 2. HAR as an AR(22)              | A5 [Solved], A6                     | B3, Wald tests                  |
| 3. Losses and weights            | A8, A10                             | A7, A9                          |
| 4. Other assets                  | B2 (ES), B4 (CL), B6 (ES, $h = 5$ ) | B1, B3, B5                      |
| 5. Multiple testing at one month | B8                                  | B7                              |
| 6. Project seed                  | C1, a level-aware kernel            | B5, Lecture 20 (kernel weights) |

- Each exercise has its own notebook section with starter code; within a pack, work in the order given
- Proposed exercises name their model: the solved exercise that uses the same technique

## A2 [Proposed]: Signature and Shuffle Identity — Task

Solution discussed in class

- ▣ **Question:** what is the signature of a three-segment path in the plane, and which channel moved first?
- ▣ **Model:** A1
- ▣ **Inputs:** A path in  $\mathbb{R}^2$  through  $(0, 0) \rightarrow (2, 1) \rightarrow (3, -1) \rightarrow (4, 2)$ , linear between points
- ▣ **Tasks**
  - (a) Write the increments and compute level 1.
  - (b) Compute level 2 with the formula of A1 (squares plus ordered pairs).
  - (c) Verify  $S^1 S^2 = S^{12} + S^{21}$  and compute the Lévy area  $A^{12}$ .
- ▣ **Report:**  $S^{(1)}$ ,  $S^{(2)}$ ,  $A^{12}$ ; **Interpretation:** which channel moved first, on balance?
- ▣ **Notebook:** section A2

## A3 [Solved]: Chen's Identity and Path Reversal — Task

- **Question:** what is the signature of a path run there and back, and why does this matter for time augmentation?
- **Inputs**
  - ▶ Path  $X : (0, 0) \rightarrow (1, 0) \rightarrow (1, 2)$  from the lecture, with  $S^{(1)} = (1, 2)$  and  $S^{(2)} = \begin{bmatrix} 1/2 & 2 \\ 0 & 2 \end{bmatrix}$
  - ▶  $\overleftarrow{X}$  is the same path run backwards:  $(1, 2) \rightarrow (1, 0) \rightarrow (0, 0)$
- **Tasks**
  - (a) Compute levels 1 and 2 of  $\text{Sig}(\overleftarrow{X})$  with the A1 formula.
  - (b) Apply Chen's identity,  $(A \otimes B)^{(2)} = A^{(2)} + A^{(1)}B^{(1)\top} + B^{(2)}$ , to compute level 2 of  $X * \overleftarrow{X}$ .
  - (c) State what the result implies for the uniqueness of the signature.
- **Report:** the two signatures and level 2 of the concatenation; **Interpretation:** why do we add a time channel in practice?
- **Notebook:** section A3

## A3 [Solved]: Solution

- (a) Reversed increments  $(0, -2), (-1, 0)$ : level 1 is  $(-1, -2) = -S^{(1)}$ ; level 2: squares  $\begin{bmatrix} 1/2 & 0 \\ 0 & 2 \end{bmatrix}$  plus the pair  $\begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix}$ , so  $\overleftarrow{S}^{(2)} = \begin{bmatrix} 1/2 & 0 \\ 2 & 2 \end{bmatrix}$
- (b) Chen:  $\begin{bmatrix} 1/2 & 2 \\ 0 & 2 \end{bmatrix} + \begin{bmatrix} -1 & -2 \\ -2 & -4 \end{bmatrix} + \begin{bmatrix} 1/2 & 0 \\ 2 & 2 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$ ; level 1 is  $S^{(1)} - S^{(1)} = 0$  as well
- (c) There and back is “tree-like”: its signature is that of the constant path; the signature identifies a path only up to such pieces [Hambly & Lyons \(2010\)](#)
- Interpretation: a strictly increasing time channel cannot go back, so tree-like pieces are impossible and the signature determines the path up to its starting point: we time-augment and supply the level separately

## A4 [Proposed]: Lévy Area of a Loop — Task

Solution discussed in class

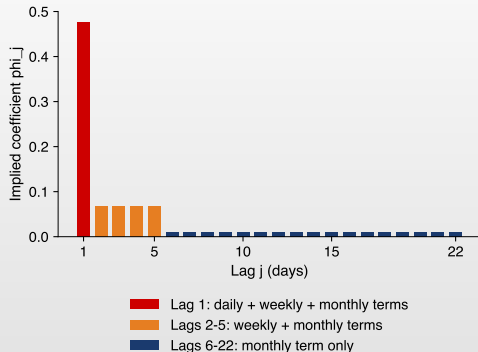
- ▣ **Question:** what does level 2 of the signature see in a closed loop that level 1 misses?
- ▣ **Model:** A3
- ▣ **Inputs:** The unit square  $(0, 0) \rightarrow (1, 0) \rightarrow (1, 1) \rightarrow (0, 1) \rightarrow (0, 0)$ , four linear segments
- ▣ **Tasks**
  - (a) Compute level 1 and the Lévy area  $A^{12}$  for the square run anticlockwise.
  - (b) Repeat for the square run clockwise, and for the square translated by  $(5, 5)$ .
  - (c) Explain why level 1 cannot distinguish the loop from a constant path while level 2 can.
- ▣ **Report:** three values of  $S^{(1)}$  and  $A^{12}$ ; **Interpretation:** which invariance does the translation illustrate?
- ▣ **Notebook:** section A4

## A5 [Solved]: HAR as a Restricted AR(22) — Task

- ▣ **Question:** which 22 lag coefficients does a HAR model for  $\ln RV$  imply for JPM?
- ▣ **Inputs**
  - ▶ VOLARE [Cipollini et al. \(2026\)](#), JPM (JPMorgan Chase), 2 Jan 2015 – 30 Jan 2026, trading days only
  - ▶ HAR with *averages of logs*:  $\ell_t = \ln RV_t$ ,  $\ell_t^{(w)} = \frac{1}{5} \sum_{j=0}^4 \ell_{t-j}$ ,  $\ell_t^{(m)} = \frac{1}{22} \sum_{j=0}^{21} \ell_{t-j}$
  - ▶ The lecture's log-HAR uses logs of averages,  $\ln RV_t^{(w)} \neq \ell_t^{(w)}$  (Jensen), and has no exact AR form
  - ▶ Full sample (2764 days), OLS, Newey–West (5 lags) s.e.:  $\hat{\beta}_d = 0.408$  (0.027),  $\hat{\beta}_w = 0.299$  (0.046),  $\hat{\beta}_m = 0.196$  (0.040)
- ▣ **Tasks**
  - (a) Substitute  $\ell_t^{(w)}$  and  $\ell_t^{(m)}$  into the HAR equation and collect the coefficient  $\phi_j$  of each lag  $\ell_{t-j+1}$ ,  $j = 1, \dots, 22$ .
  - (b) Compute  $\phi_1, \phi_2 = \dots = \phi_5$  and  $\phi_6 = \dots = \phi_{22}$  from the JPM estimates and plot them.
  - (c) Compute the sum  $\sum_{j=1}^{22} \phi_j$ .
- ▣ **Report:** the three distinct values, the sum, the chart; **Interpretation:** what does the sum measure, and is the process stationary?
- ▣ Notebook: section A5

## A5 [Solved]: Solution

- (a)  $\beta_w \ell_t^{(w)} = \sum_{j=1}^5 (\beta_w/5) \ell_{t-j+1}$  and  $\beta_m \ell_t^{(m)} = \sum_{j=1}^{22} (\beta_m/22) \ell_{t-j+1}$ : lag 1 collects all three terms, lags 2–5 the weekly and monthly ones, lags 6–22 only the monthly one
- (b)  $\phi_1 = 0.408 + 0.299/5 + 0.196/22 = 0.476$ ;  $\phi_{2..5} = 0.0688$ ;  $\phi_{6..22} = 0.0089$
- (c)  $\sum_j \phi_j = \beta_d + \beta_w + \beta_m = 0.903$
- Interpretation: the sum is the persistence of log RV; all  $\phi_j \geq 0$  and the sum is below 1, so the process is stationary, with a slow, long-memory-like decay



## A6 [Proposed]: Testing the HAR Restrictions — Task

Solution discussed in class

- ▣ **Question:** how do we test the restrictions that HAR imposes on an unrestricted AR(22)?
- ▣ **Model:** A5
- ▣ **Inputs**
  - ▶ The unrestricted AR(22) for  $\ell_t = \ln RV_t$  has an intercept and slopes  $\phi_1, \dots, \phi_{22}$ ; HAR with averages of logs (A5) is its restricted version
  - ▶  $\phi$  is the linear projection of  $\ell_{t+1}$  on its 22 lags; weakly dependent data with finite fourth moments; errors may be heteroskedastic and serially correlated
- ▣ **Tasks**
  - (a) Count the linear restrictions and write them as  $R\phi = 0$ .
  - (b) Write the Wald statistic with an HAC covariance and its distribution under  $H_0$ .
  - (c) State the 5% critical value.
- ▣ **Interpretation:** if the test rejects in a long sample, should we drop HAR for forecasting?
- ▣ **Notebook:** section A6

## A8 [Proposed]: QLIKE With Its Arguments Swapped — Task

Solution discussed in class

- ▣ **Question:** which forecasts does a QLIKE with swapped arguments reward?
- ▣ **Model:** A7
- ▣ **Inputs:** A program computes  $L(RV, F) = F/RV - \ln(F/RV) - 1$ , QLIKE with the arguments swapped
- ▣ **Tasks**
  - (a) Find the forecast  $F^\dagger$  that minimises  $\mathbb{E}_t[L(RV, F)]$ .
  - (b) Compare  $F^\dagger$  with  $\mathbb{E}_t[RV]$  using Jensen's inequality.
  - (c) Say which models the swapped loss favours.
- ▣ **Interpretation:** why can this bug pass elementary checks?
- ▣ **Notebook:** section A8

## A10 [Proposed]: Limits of the Temperature — Task

Solution discussed in class

- ▣ **Question:** what happens to the kernel weights at the two extreme temperatures, and why scale by the median distance?
- ▣ **Model:** A9
- ▣ **Inputs:**  $n$  past windows, distances  $\delta_\tau > 0$ , temperature  $\gamma = c/\text{median}(\delta)$  as in the lecture
- ▣ **Tasks**
  - (a) Find the limits of  $w$  and  $n_{\text{eff}}$  when  $c \rightarrow 0$ .
  - (b) Find the limits when  $c \rightarrow \infty$  (assume a unique nearest window).
  - (c) Show that multiplying all distances by  $k > 0$  leaves the weights unchanged.
- ▣ **Interpretation:** why is the median scaling needed to compare  $c$  across 50 assets?
- ▣ Notebook: section A10

## B2 [Proposed]: The Lévy Area for E-mini S&P 500 Futures — Task

Solution discussed in class

- ▣ **Question:** does the Lévy area predict next month's volatility of E-mini S&P 500 futures (ES)?
- ▣ **Model:** B1
- ▣ **Inputs:** VOLARE, futures file, 28 Sep 2009 – 30 Jan 2026, ES;  $rv5$ , open and close prices; target, regressors and 22 Newey–West lags as in B1
- ▣ **Tasks**
  - (a) Repeat steps (a)–(d) of B1 for ES.
  - (b) Test the coefficient of  $A_t$  at 1% and 5%, two-sided.
  - (c) Compare the  $R^2$  gain with the one for JPM.
- ▣ **Interpretation:** what does a positive coefficient say about windows with a negative area, and can the sign alone label them leverage episodes?
- ▣ **Notebook:** section B2

## B4 [Proposed]: HAR, SHAR and HARQ for Crude Oil — Task

Solution discussed in class

- ▣ **Question:** do the SHAR asymmetry and the HARQ effect hold for crude oil futures (CL)?
- ▣ **Model:** B3
- ▣ **Inputs:** VOLARE, futures file, 28 Sep 2009 – 30 Jan 2026, CL; rv5, rsp5, rsn5, rq5; models, tests and Newey–West (5 lags) as in B3
- ▣ **Tasks**
  - (a) Estimate SHAR and test  $\beta^- > \beta^+$  as in B3(b).
  - (b) Estimate HARQ and test  $\beta_Q < 0$  as in B3(c); plot the coefficients with 95% HAC intervals.
  - (c) Compare both results with JPM.
- ▣ **Interpretation:** why might the asymmetry be stronger for a commodity than for a bank stock?
- ▣ **Notebook:** section B4

## B6 [Proposed]: Sig-LK vs log-HAR for ES, One Week Ahead — Task

Solution discussed in class

- ▣ **Question:** does Sig-LK beat log-HAR for ES when the target is the average RV of the next five days?
- ▣ **Model:** B5
- ▣ **Inputs:** VOLARE, futures file, 28 Sep 2009 – 30 Jan 2026, ES; design of B5 with  $h = 5$  and  $c = 1.0$  (futures validation block)
- ▣ **Tasks**
  - (a) Produce weekly forecasts of Sig-LK and log-HAR with the design of B5.
  - (b) Compute the QLIKE ratio and its stationary-bootstrap 95% CI (mean block length 10).
  - (c) Run the DM test (bandwidth at least  $h = 5$ ) and the conditional GW test; plot the cumulative loss difference.
- ▣ **Interpretation:** why is the CI so asymmetric, and what does it imply for the power of DM?
- ▣ **Notebook:** section B6

## B8 [Proposed]: Multiple Testing at One Month Against HAR — Task

Solution discussed in class

- ▣ **Question:** on how many assets does Sig-LK beat HAR one month ahead after correcting for 50 tests?
- ▣ **Model:** B7
- ▣ **Inputs:** Lecture 20 results: 50 one-sided DM  $p$ -values for “Sig-LK better than HAR” at  $h = 22$  (QLIKE)
- ▣ **Tasks**
  - (a) Count the rejections at 5% without correction.
  - (b) Apply Holm and BH as in B7, count the rejections and name the asset with the smallest  $p$ -value.
  - (c) Compare with the pooled mean QLIKE ratio of Sig-LK relative to HAR at  $h = 22$  (1.334).
- ▣ **Interpretation:** how can Sig-LK beat HAR on several assets and still lose on average?
- ▣ **Notebook:** section B8

## C1 [Proposed]: A Level-Aware Signature Kernel — Task

Solution discussed in class

- ▣ **Question:** does a kernel that also sees the *level* of volatility make the weights useful in a crash?
- ▣ **Inputs**
  - ▶ The kernel of the lecture is translation-invariant, so it cannot recognise a high-volatility regime (Lecture 20, weights on 16 March 2020); data: VOLARE, the 40 stocks, 2015–2026
  - ▶ Idea: prepend a **basepoint** [Morrill et al. \(2020\)](#): the path starts at the training mean of  $\ln RV$  and jumps to the first value of the window; by Chen,  $\text{Sig} = \exp_{\otimes}(v) \otimes \text{Sig}(X)$  with  $v$  the standardised level jump
- ▣ **Tasks**
  - (a) Write a one-page pre-registration: 40 stocks,  $h = 1$ , the lecture design,  $c = 0.5$ , QLIKE, DM with Holm, the COVID-19 subperiod reported separately.
  - (b) Implement the basepoint kernel (use the Chen update of the notebook) with the same features as Sig-LK, and forecast.
  - (c) Compare with Sig-LK and log-HAR: mean QLIKE ratios (all days and COVID-19), wins over log-HAR raw and after Holm.
- ▣ **Interpretation:** if seeing the level does not help, where would you look next (features, target, horizon)?
- ▣ Notebook: section C1

## Further Study on Quantinar

### Advanced courses on quantinar.com

- ▣  Statistics of Financial Markets
- ▣  Applied Time Series Analysis with Python

### Code and data

- ▣ Seminar notebook (Colab); Quantlets: [github.com/danpele/MFM/tree/main/Quantlets/Ch\\_20](https://github.com/danpele/MFM/tree/main/Quantlets/Ch_20); data: [volare.unime.it](https://volare.unime.it)

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